Groundwork Aviation

THE FREE FIELD GUIDE · NO. 01

Decode Any *METAR & TAF*

The 20-minute guide that turns the scariest string of letters in aviation into something you can read at a glance — and explain at your checkride.

Written & vetted by a working airline pilot

8,000+ hours on the flight deck

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START HERE

Why this one skill is worth your evening

Weather interpretation is the single most-missed area on the private pilot knowledge test. Not because it's the hardest — because most people learn it as memorised trivia instead of a system.

Here's the good news, and the entire premise of this guide: a METAR isn't a code to memorise. It's a *sentence with a fixed word order*. Once you know the order and what each slot means, you can read any report in the world — Sydney, San Francisco, or a field you've never heard of. Same with a TAF, which is just a METAR that talks about the future.

We'll do four things: walk the word order once, decode a real report slot by slot, do the same for a TAF, then show you the six traps that catch people on the exam. By the last page you'll have a one-page cheat sheet to keep on your kneeboard.

> ★ THE MINDSET SHIFT
>
> Stop reading a METAR left-to-right like a puzzle. Read it as answers to fixed questions, always asked in the same order: **Who? When? Wind? How far can I see? What's blocking the sky? How warm? What's the pressure?**

THE ANATOMY

The fixed word order of every METAR

THE SLOT	THE QUESTION IT ANSWERS
Station	Which airport is this report for? (a 4-letter ICAO identifier)
Date / Time	When was it issued? Always in UTC — “Zulu” time.
Wind	Which direction is it from, and how strong? Gusts noted here too.
Visibility	How far can you see horizontally?
Weather	Rain, mist, thunder? (Often absent — that's good news.)
Sky / Clouds	How much cloud, and at what height above the field?
Temp / Dewpoint	How warm is it, and how close is the air to saturation?
Altimeter	What pressure do you set so your altimeter reads true?
Remarks (RMK)	Anything extra the station wants to flag.

Memorise the *order*, not the contents. The order never changes — so even an unfamiliar group is just “the next answer in the sentence.”

WORKED EXAMPLE

One real report, decoded slot by slot

Here's an actual-format METAR. Don't panic at the wall of characters — we'll take it one slot at a time, in order.

```
// THE RAW REPORT
KSFO 301853Z 28012G20KT 10SM FEW015 SCT200 18/12 A3001
RMK AO2 SLP163
```

GROUP	WHAT IT MEANS
KSFO	**San Francisco International.** In the US, the leading “K” is just a region prefix — the airport is “SFO”.
301853Z	**Issued the 30th, at 18:53 UTC.** The “Z” means Zulu/UTC. Always convert to your local time mentally.
28012G20KT	**Wind from 280° true, at 12 knots, gusting 20.** Direction is always 3 digits; “G” flags gusts.
10SM	**Visibility 10 statute miles** — the maximum normally reported. Clear day.
FEW015	**A few clouds at 1,500 ft** above the field. Heights are in hundreds of feet AGL: “015” → 1,500.
SCT200	**Scattered clouds at 20,000 ft.** High and harmless here.
18/12	**Temperature 18°C, dewpoint 12°C.** The gap (the “spread”) tells you how close the air is to forming cloud or fog.
A3001	**Altimeter 30.01 inHg.** The “A” = altimeter in inches of mercury. Set this and your altimeter reads true.
RMK AO2 SLP163	**Remarks:** automated station with a precipitation sensor (AO2); sea-level pressure 1016.3 hPa (SLP163).

> READ IT IN ONE BREATH
>
> “San Francisco, 18:53 Zulu, wind 280 at 12 gusting 20, ten miles, few at 1,500 and scattered at 20,000, eighteen over twelve, altimeter three-zero-zero-one.” That's it. You just read a METAR like a pilot.

THE TWO KEYS YOU ACTUALLY MEMORISE

Sky cover & common weather codes

SKY COVER	MEANS	WEATHER	MEANS
SKC / CLR	Sky clear	RA	Rain
FEW	Few (1–2 eighths)	SN	Snow
SCT	Scattered (3–4)	BR	Mist
BKN	Broken (5–7)	FG	Fog
OVC	Overcast (8)	TS	Thunderstorm

Intensity prefixes: - light, (none) moderate, + heavy. So +TSRA = heavy thunderstorm with rain. BKN or OVC is your *ceiling* — the lowest broken/overcast layer — which drives a lot of your go/no-go.

THE FORECAST COUSIN

A TAF is just a METAR about the future

A **TAF** (Terminal Aerodrome Forecast) uses the same vocabulary you just learned, with a few time-based words bolted on. Same slots, same codes — it just describes what's *expected* over a period (usually 24–30 hours) rather than what's happening now.

```
// THE RAW FORECAST
TAF KSFO 301730Z 3018/3124 28010KT 9SM FEW020
FM010200 30008KT 6SM BR BKN012
TEMPO 0104/0108 3SM OVC008
```

NEW WORD	WHAT IT MEANS
3018/3124	**Valid period:** from the 30th 18:00Z to the 31st (“31” → shown as 01 day rolls) 24:00Z. Read as day/hour to day/hour.
FM010200	**FROM the 1st at 02:00Z**, conditions change to what follows — a rapid, lasting change. Everything after replaces the previous line.
TEMPO 0104/0108	**TEMPORARILY** between the 1st 04:00Z and 08:00Z, expect this — brief fluctuations lasting under an hour each.
BECMG	(Not shown) **BECOMING** — a gradual transition over a stated window.
PROB30/40	(Not shown) a **30% / 40% probability** of the conditions that follow.

EXAM SURVIVAL

The 6 traps that catch people

1. **Zulu vs. local time**

   Every time in a METAR/TAF is UTC. Questions love to make you convert to local — know your offset cold, and remember it shifts with daylight saving.

2. **Cloud height is AGL, in hundreds of feet**

   `BKN012` is 1,200 ft *above the airfield*, not 12 ft and not above sea level. Drop the last two zeros: append “00”.

3. **Ceiling = lowest BKN or OVC**

   FEW and SCT layers do *not* count as a ceiling. The exam tests whether you know the ceiling is the first broken or overcast layer you meet going up.

4. **Wind direction is TRUE, not magnetic**

   METAR/TAF winds are referenced to true north. The winds ATIS or the tower reads to you over the radio are magnetic. Same wind, different reference — a classic gotcha.

5. **A small temp/dewpoint spread means fog risk**

   When temperature and dewpoint converge (e.g. 12/11), the air is near saturation — watch for mist, fog and a dropping ceiling. The numbers are a warning, not just data.

6. **FM resets the forecast; TEMPO/PROB don't**

   `FM` is a clean break — everything after it replaces what came before. `TEMPO` and `PROB` are temporary overlays on the prevailing forecast, not permanent changes.

> ★ CHECKRIDE-ORAL UPGRADE
>
> The written asks you to *recognise* these. The examiner will ask you to *explain* them — “why does a narrow temp/dewpoint spread matter to you today?” Practice saying the **why** out loud, not just the definition. That's the difference between passing the test and being trusted with an aircraft.

REPS

Decode these yourself, then check

Cover the grey answer, read the report aloud, then reveal. Do all four and the skill is yours.

```
METAR KBOS 121651Z 04015KT 1/2SM +RA BR OVC004 09/09 A2978
```

Boston, 16:51Z. Wind 040°/15kt. Visibility ½ mile in heavy rain and mist. Overcast at 400 ft (that's your ceiling — low). 9°C / 9°C — zero spread, fully saturated. Altimeter 29.78. Translation: ugly, wet, low-ceiling day. Not VFR.

```
METAR KSEA 120753Z 00000KT 10SM CLR 08/07 A3015
```

Seattle, 07:53Z. Wind calm (`00000KT`). 10 miles visibility. Sky clear (`CLR` — no clouds detected below 12,000 ft by the automated station). 8°C / 7°C — only a 1° spread. Altimeter 30.15. A fine-looking report — but clock that tight temperature/dewpoint spread: it's prime fog or low-cloud weather, especially around dawn.

```
METAR KDEN 121553Z 36021G31KT 10SM FEW070 SCT110 28/M01 A3012
```

Denver, 15:53Z. Wind 360°/21kt gusting 31 — strong and gusty. 10 miles vis. Few at 7,000, scattered at 11,000. 28°C, dewpoint MINUS 1°C ("M" = minus). Altimeter 30.12. Big spread, high cloud — but note the gusts and high field elevation: think density altitude.

```
TAF KLAX 121120Z 1212/1318 25010KT 6SM BR SCT008 FM121700 26012KT 10SM SKC
```

LA forecast issued 11:20Z, valid 12th 12:00Z → 13th 18:00Z. Initially wind 250°/10kt, 6 miles in mist, scattered at 800 ft. FROM the 12th 17:00Z: wind 260°/12kt, 10 miles, sky clear. Morning marine layer burning off by late morning — textbook coastal pattern.

> IF THOSE CLICKED
>
> You now read weather better than a lot of people walking into their checkride. That's one topic down — and proof of how we teach the rest: the system first, the memorising second.

PRINT THIS PAGE · KEEP IT ON YOUR KNEEBOARD

METAR / TAF Quick Decoder

WORD ORDER (METAR)

1 Station	4-letter ICAO
2 Time	DDHHMMZ (UTC)
3 Wind	dddss(Gss)KT — true
4 Visibility	SM (US) / metres
5 Weather	RA/SN/BR/FG/TS
6 Sky	FEW/SCT/BKN/OVC + hhh
7 Temp/Dew	TT/DD (M = minus)
8 Altimeter	Annnn / Qnnnn

SKY COVER

SKC/CLR	Clear
FEW	1–2 eighths
SCT	3–4 eighths
BKN	5–7 · ceiling
OVC	8 · ceiling

INTENSITY

- / (none) / +	light / mod / heavy

TAF TIME-WORDS

DDHH/DDHH	Valid period
FM	From — clean reset
BECMG	Becoming — gradual
TEMPO	Temporary <1hr
PROB30/40	% chance

DON'T GET CAUGHT

Time	Always UTC / Zulu
Cloud ht	×100 ft AGL
Ceiling	Lowest BKN/OVC
Wind	TRUE in text, MAG on radio
Spread	Narrow → fog risk
Pressure	A = inHg · Q = hPa

Read every slot in order, every time. The string stops being scary the moment it becomes a sentence.

Flying internationally (ICAO)? *Visibility is in metres, pressure is `Q` in hPa (not `A` in inHg), and you'll meet `CAVOK` — visibility 10 km+, no cloud below 5,000 ft (or the highest minimum sector altitude), no CB/TCU, and no significant weather. US reports don't use CAVOK.*

US remarks (RMK) *can also carry precipitation totals — e.g. `P0009` (0.09 in in the last hour), with separate 6-hour and 24-hour groups.*

THAT'S ONE TOPIC MASTERED

You just learned the thing that fails the most people. There's more where that came from.

Groundwork Aviation breaks down the topics flight school rushes — airspace, performance, aerodynamics, and the checkride oral — in the same plain-English, system-first way. Written and vetted by a working airline pilot, so the *why* always holds up.

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COMING NEXT FROM GROUNDWORK

→ **The Checkride Oral Playbook** — answer like a pilot, not a flashcard

→ **Airspace, finally clear** — read any sectional with confidence

→ **Performance & W&B without the panic** — density altitude, demystified